MIL-G-46690E(EA)
23 March 1976
SUPERSEDING
MIL-G-46690D(MU)
7 October 1968

MILITARY SPECIFICATION

GRENADE, HAND-RIFLE, SMOKE, WP, M34

FILLING AND ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the filling and assembly of a bursting-type grenade.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-C-215 - Chemical Agent, White Phosphorus (WP).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1168 - Lot Numbering of Ammunition.

FSC 1330

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DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

DL13-7-4 - Grenade, Hand-Rifle, Smoke, WP, M34.
A18-60-966 - Filling Identification.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D13-7-4 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the grenade shall conform to the specifications and drawings listed on Drawing D13-7-4 and subsidiary drawings.

3.2 Filling and assembly.

3.2.1 Filling. The grenade shall be clean and free of foreign matter and shall be filled with molten WP conforming to MIL-C-215 (see 6.2 and 6.3) at a temperature of 140° ± 10°F. When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0 After test for leakage, the WP shall be solidified with the grenade in a filling-end-up position.

3.2.2 Filling height. There shall be a 7 to 15 percent void by total volume existing above the surface of the water over the WP in the filled and closed grenade. The depth of water remaining above the WP in the filled and closed grenade shall not exceed 1/8 inch. The grenade shall be radiographed in a vertical, filling-end-up position while the temperature of the WP is 130° ± 10°F when tested as specified in 4.2.4.1, to determine filling height.

3.2.3 Assembly. The grenade shall be assembled as shown on Drawing D13-7-4.

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3.3 Cleaning after closing. The exterior surface of the filled and closed grenade and the interior of the burster casing shall be thoroughly cleaned to remove any grease, oil, WP or other foreign matter and thoroughly dried prior to inserting the fuze (see 6.4).

3.4 Leakage. The closed and cleaned grenade shall not leak when maintained at a temperature of 160° ± 5°F for a period of 8 hours. Alternate method: the grenade shall not leak when maintained at a temperature of 207° to 212°F for a minimum period of 15 minutes. The grenade shall be tested as specified in 4.2.4.2.

3.5 Functioning. After release of the fuze lever, the grenade shall burst within 4.1 to 5.2 seconds and disperse the majority of the WP when tested in accordance with 4.2.4.3.

3.6 Workmanship. The grenade shall be free of all foreign matter (oil, grease, and contamination) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each lot shall be identified and controlled in accordance with MIL-STD-1168. However, no more than one lot of each component (bodies, burster wells and fuzes) shall be contained in any one lot of assembled grenades.

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4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive test. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample grenades shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.2.3.2 For destructive test. Sample grenades shall be tested in accordance with 4.2.4.3. For grenade failure, and AQL of 1.5 percent defective shall be used. A single delay time less than 3 seconds shall reject the lot.

4.2.3.3 For critical defects. Each grenade in the lot shall be inspected for its critical characteristics listed in the classification of defects.

4.2.3.4 Classification of defects.

(a) Grenade, hand-rifle, smoke, WP, M34 (Dwg D13-7-4).

Categories	Defects	Acceptance standards
Critical:		
1	WP leakage	4.2.4.2
2	WP contamination (3.3)	
Major:	AQL 1.0 percent defective	
101	Filling excessive	4.2.4.1
102	Filling insufficient	4.2.4.1
103	Component missing	
104	Component incorrectly assembled	
105	Excess water in grenade	4.2.4.1
106	Marking incorrect	
107	Torque incorrect	
108	Workmanship (3.6)	

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Minor:	AQL 2.5 percent defective
201	Protective finish missing
202	Marking missing
203	Rust inhibitor missing

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Container leakage
102	Packing wood not preserved and dried
103	Loose pack
104	Instruction card missing
105	Marking incorrect
106	Marking missing
Minor:	AQL 10.0 percent defective
201	Marking illegible

4.2.4 Tests.

4.2.4.1 Filling height. The filled and closed grenade shall be radiographed while the WP is molten as specified with the grenade in a vertical filling-end-up position.

4.2.4.2 Leakage. After final cleaning, the unfuzed grenades shall be placed in an oven in a vertical, filling-end-down position for the 160° ± 5° F method (form the alternate method at 207° to 212° F, they shall be placed in a filling-end-up position) and maintained at the specified temperature for the required period of time, and observed for leakage. When tested at 207° to 212° F, the munition temperature shall be measured by thermocouples and/or thermometers inserted in the burster wells of the test items at random locations. The emission of smoke constitutes leakage.

4.2.4.3 Functioning. The grenade shall be fired by using either a remote control firing device to withdraw the safety pin, or a grenade launcher provided with a barricade (see 6.2).

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5. PREPARATION FOR DELIVERY

5.1 Packaging, level A. Grenades shall be individually packaged as specified on Drawing D13-9-98. Two keys, Drawing D22-129-1, shall be packaged together in accordance with method IC-1 or IC-3 of MIL-P-116.

5.2 Packing, level A. The grenades, packaged in accordance with 5.1, shall be packed, 16 per box, in boxes conforming to style 4, Class 2, Grade A, 50-100 lb limit for type 1 or 2 load for overseas shipment, PPP-B-621, having inside dimensions in inches of 11 1/2 by 11 1/2 by 6 3/4 length, width and depth respectively. Filler pads fabricated of fiberboard conforming to either Grade W5c or W6c, PPP-F-320 shall be included to cover all inner faces of the box to separate the packaged grenades from the box. Two packaged keys and one instruction card conforming to Drawing C13-7-1 shall be included in each box and placed on top of the packaged grenades. Boxes shall be closed and strapped in accordance with the appendix to PPP-B-621, including additional filler pads of type cited above, as necessary to restrict free movement of the packaged grenades. Strapping shall conform to either type I or IV, Class A or B, QQ-S-781.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129 and Drawing A18-60-966. Shipping containers shall be marked in accordance with Drawing D13-7-22.

5.4 Palletization. Shipping containers marked in accordance with 5.3 shall be palletized and then marked in accordance with Drawing D13-7-21.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 Caution. Due to the hazards inherent in the handling of WP, suppliers are cautioned to protect personnel from injury while filling and testing grenades.

6.3 Filling aid. Grenades filled with approximately 0.75 pounds of WP have been found to comply with 3.2 provided the internal configuration of the grenades are within the tolerances specified on the drawings. This information is furnished as an aid to procurement and not as a criterion for filling.

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6.4 Cleaning. The filled grenade may be cleaned by passing through a hot water and detergent spray washer or by flushing with hot water while vigorously scrubbing with steel wool. Additional cleaning of the burster well may be accomplished with a steam lance and a power driven absorbent swab.

6.5 Refuzing. If failure of the grenade is caused by the fuze, refuzing is permitted.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1330-A118

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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